

ANOVA: Degrees of freedom

→ think in terms of number of deviations we have squared

$$SS_{\text{total}} = \sum \sum (y_{ij} - \bar{y})^2$$

→ N deviations around one estimated grand mean squared

→ to estimate total SS; use one estimate = lose one degree of freedom

$$\therefore df_{\text{total}} = N - 1$$

$$SS_{\text{treat}} = \sum \sum (\bar{y}_i - \bar{y})^2$$

→ sum of deviations of k groups around the estimated grand mean

→ to estimate SS_{treat} ; use one estimate = lose one degree of freedom

$$\therefore df_{\text{treat}} = k - 1$$

$$SS_{\text{resid}} = \sum \sum (y_{ij} - \bar{y}_i)^2$$

→ k sets of n deviations around one estimate

→ to estimate SS_{resid} ; use k estimates = lose k degrees of freedom $\therefore k(n-1)$